

Highlights

From Detection to Maintenance Action: Streaming Sensor-Fault Typing for Self-Supervised Anomaly Detection in Industrial Systems

Marek Wadinger, Michal Kvasnica

- Streaming detector attributes each flagged sensor to a maintenance-relevant fault type
- Reaches 90% real-fault detection at 7.4% healthy false-positive rate on battery-depletion faults
- $O(1)$ -memory conditional-residual typing composed with a self-supervised online detector
- Operating envelope characterized: bias absorption, dead-band, and correlation limits made explicit
- Online adaptation is mandatory; a static prefix model false-alarms at 0.97 on a diurnal stream

From Detection to Maintenance Action: Streaming Sensor-Fault Typing for Self-Supervised Anomaly Detection in Industrial Systems

Marek Wadinger^{a,*}, Michal Kvasnica^a

^a*Institute of Information Engineering, Automation and Mathematics, Slovak University of Technology in Bratislava, Radlinskeho 9, 812 37, Bratislava, Slovakia*

Abstract

Streaming anomaly detectors tell operators *that* a sensor reading is abnormal and, when interpretable, *which* signal and in *which* direction. They do not tell operators *what to do*: whether to recalibrate, replace, or clean a sensor, or to investigate the process itself. We close this gap with a streaming sensor-fault classifier that attributes each flagged signal to a maintenance-relevant fault type—bias, drift, accuracy loss, or freezing—directly from the conditional residuals of a self-supervised online detector, using $O(1)$ memory and no history buffers. On real battery-depletion faults in the Intel Berkeley Lab deployment, the composed system reaches 90% real-fault detection at a 7.4% healthy false-positive rate, and types the resulting faults predominantly as freezing (0.93), the signature of sensors stuck at degraded values. We further characterize the operating envelope on controlled injections: freezing and drift are recovered with recall 1.000 and 0.741, while bias recall is monotone in conditional standard deviation—the interpretable conditional model absorbs constant offsets on strongly coupled signals (recall 0.059) and isolates them on weakly coupled ones (0.941). We show that two original claims do not survive validation: freezing is not a detection blind spot of the base detector (it catches every injected freeze, usually faster than the dedicated test), and adaptation robustness comes from graded change-point suppression rather than residual cancellation. The result is a deployable typing layer with a

**Phone numbers:* +421 902 810 324 (Marek Wadinger)

Email addresses: `marek.wadinger@stuba.sk` (Marek Wadinger),
`micHAL.kvasnica@stuba.sk` (Michal Kvasnica)

URL: `uiam.sk/~wadinger` (Marek Wadinger)

rigorously mapped—rather than overstated—operating envelope.

Keywords: Fault diagnosis, Sensor fault classification, Online learning, Anomaly detection, Self-supervised learning, Streaming algorithms

1. Introduction

In industrial monitoring, the cost of a sensor fault is paid not when it is detected but when it is acted upon. A reading that drifts away from its true value, a transmitter stuck at its last sample, or an instrument whose noise floor has grown all demand different maintenance responses: recalibration, replacement, cleaning, or process investigation. An operator who learns only that a signal is anomalous, and even which signal and in which direction, must still diagnose the failure mode before dispatching the correct action. The fault *type* is what drives the maintenance action, and it is precisely what most streaming detectors leave to human triage.

The base detector we extend is a self-supervised, online anomaly detector for SCADA-based industrial systems (Wadinger and Kvasnica, 2023). It scores each signal by the conditional cumulative probability of its current value given the remaining signals, flags a value when that probability falls below α or above $1 - \alpha$ (with $\alpha \approx 0.00133$, i.e. roughly three conditional standard deviations), and adapts its conditional model online with change-point awareness. Crucially, it is interpretable: it isolates the responsible signal and the direction of the excursion, and exposes per-signal operating limits. What it does not provide is the fault *type*—it localizes the anomaly but does not characterize the failure mode, leaving the detection-to-action gap open.

We close that gap with a streaming sensor-fault classifier that attributes each flagged signal to one of four maintenance-relevant types—bias, drift, accuracy loss, or freezing—computed from the same conditional residuals the base detector already produces. The classifier maintains only a small set of exponentially weighted running statistics per signal: it uses $O(1)$ memory, keeps no history buffers, and adds negligible cost to the streaming pipeline. The fault taxonomy itself is not new; we adopt the bias / noise / constant / drift categories established for sensor faults in wireless and environmental sensing (Sharma et al., 2010; Ni et al., 2009). Our contribution is the *streaming conditional-residual realization* of this taxonomy, composed directly with a self-supervised detector so that type attribution rides on the same online statistics as detection.

We lead with the practical outcome. On real battery-depletion faults from the Intel Berkeley Lab deployment (Bodik et al., 2004)—where a mote’s failing battery degrades its temperature and humidity readings to physically impossible values—the composed system reaches 90% real-fault detection at a 7.4% healthy false-positive rate, and types the resulting faults predominantly as freezing (0.93), the expected signature of a transmitter stuck at a degraded value. Online adaptation is not optional here: a static model fit on a short prefix of the same diurnal stream false-alarms at a rate of 0.97. The typing layer thus converts an adaptive detector’s alarm into an actionable maintenance label on a real fault population.

We then characterize the operating envelope honestly, treating it as support for—not a qualification of—the practical result. Two claims we initially expected to hold do not survive our own validation, and we report them as refuted. First, freezing is *not* a detection blind spot of the base detector: it catches every injected freeze on every channel, usually faster than a dedicated fixed-window freeze test, so the freeze test contributes type *attribution* (a stuck sensor versus a process excursion), not detection capability. Second, the robustness of the adaptive detector to faults that arrive during adaptation comes from graded change-point suppression, not from conditional-residual cancellation. We treat this honesty as a feature: the envelope is mapped rather than overstated, and the limitations—bias absorption on strongly coupled signals, the limited cancellation of a two-signal conditional model, and the masking dead-band introduced by suppression—are quantified and explained.

The contributions of this paper are:

1. A streaming, $O(1)$ -memory sensor-fault classifier that types each flagged signal as normal, bias, drift, accuracy loss, or freezing from the conditional residuals of a self-supervised online detector, with no history buffers (Section 3).
2. A demonstration on real battery-depletion faults that the composed system reaches 90% real-fault detection at a 7.4% healthy false-positive rate and assigns maintenance-relevant types, with online adaptation shown to be mandatory (Section 5).
3. A rigorous characterization of the operating envelope on controlled injections, including the monotone dependence of bias recall on conditional coupling, and two refuted claims (freezing as a blind spot; residual cancellation as the source of adaptation robustness) reported as

70 such (Sections 5 and 6).

71 2. Related Work

72 2.1. The base detector

73 The detector we extend scores anomalies in SCADA-based industrial sys-
74 tems by self-supervised online learning (Wadinger and Kvasnica, 2023). For
75 each signal it evaluates the conditional cumulative distribution of the current
76 value given the remaining signals under a multivariate Gaussian model, flags
77 the value when this probability lies outside $[\alpha, 1 - \alpha]$ with $\alpha \approx 0.00133$, and
78 updates the conditional model online while remaining aware of change points
79 so that legitimate regime shifts are absorbed rather than alarmed. The de-
80 tector is interpretable: it identifies the responsible signal, the direction of the
81 excursion, and exposes per-signal operating limits. It localizes anomalies but
82 does not classify the failure mode, which is the gap this work addresses. We
83 reuse its conditional-residual stream as the substrate for typing rather than
84 introducing a separate diagnostic pipeline.

85 2.2. Sensor-fault taxonomies

86 The categories we attribute—bias, drift, accuracy loss (noise), and freez-
87 ing (constant)—are standard in the sensor-fault literature rather than novel
88 to this work. Sharma et al. (2010) characterize the recurrent fault signatures
89 of deployed sensors, including offset (bias), gain, and stuck-at faults, and
90 study detection methods against them. Ni et al. (2009) likewise organize
91 sensor data faults into a taxonomy spanning calibration offset, drift, and
92 constant (frozen) readings. We adopt this established taxonomy unchanged;
93 our contribution is its streaming, conditional-residual realization, not the
94 categories themselves.

95 2.3. Fault-injection protocols

96 Because no public stream simultaneously labels all four fault types per
97 signal and per timestep, controlled injection is the established route to per-
98 type evaluation. Lai et al. (2021) introduce a benchmark methodology that
99 injects parameterized anomaly patterns into time series to evaluate detectors
100 under known ground truth. We follow this precedent, injecting the four
101 typed faults into a multivariate process at controlled magnitudes and onsets
102 so that recall and confusion can be measured per type, and we triangulate the
103 synthetic findings against the real-fault deployment described in Section 4.

104 2.4. Offline fault classifiers

105 A large body of work classifies sensor and process faults offline, typically
106 by training supervised or windowed models on labelled fault episodes and
107 extracting features over batches of history. Such methods can exploit rich
108 temporal context but require stored windows, labelled fault data, and a train-
109 ing phase, which conflicts with the streaming, self-supervised, and label-free
110 setting of the base detector. Our classifier instead derives the type from the
111 running conditional residuals already maintained by the detector, in $O(1)$
112 memory and without history buffers, so that typing is available at the same
113 latency and operational footprint as detection.

114 3. Method

115 3.1. Conditional residuals as the typing substrate

116 The base detector (Wadinger and Kvasnica, 2023) maintains, for each
117 signal i , a conditional model of its current value x_i given the remaining
118 signals x_{-i} . We define the conditional residual

$$r_i = x_i - \mu_{\text{cond},i}(x_{-i}), \quad (1)$$

119 the difference between the observed value and its conditional mean, and the
120 conditional standard deviation $\sigma_{\text{cond},i}$ already used by the detector to form its
121 score. Both quantities are available at every step at no additional modelling
122 cost. The classifier operates entirely on the stream of residuals $\{r_i\}$ and their
123 conditional scale $\sigma_{\text{cond},i}$: it never stores past samples and never refits a model
124 over a window. This keeps the per-signal state to a fixed set of scalars and
125 the per-step cost constant, matching the streaming, $O(1)$ -memory footprint
126 of the detector it extends.

127 3.2. Running statistics

128 For each signal the classifier maintains a small set of exponentially weighted
129 moving (EWM) statistics over the residual stream. We track a fast (short
130 half-life) and a slow (long half-life) EWM mean of r_i , denoted $m_{\text{fast},i}$ and
131 $m_{\text{slow},i}$, and an EWM variance v_i of r_i . We additionally track the EWM
132 variance of the raw signal increments to detect a stalled transmitter. All up-
133 dates are the standard constant-memory recursions; no buffer of past values
134 is required. The half-lives are the only temporal hyperparameters of the typ-
135 ing layer and are shared across signals unless per-signal calibration is applied
136 (Section 6).

137 3.3. The four type tests

138 Given the running statistics, a flagged signal is attributed to a type by
139 the following tests, each expressed in units of the conditional scale so that
140 thresholds are dimensionless.

141 *Bias..* A persistent offset of the residual from zero, with stable variance,
142 indicates a constant calibration error. We declare *bias* when the slow residual
143 mean is large in conditional-scale units, $|m_{\text{slow},i}|/\sigma_{\text{cond},i}$ exceeding a threshold,
144 while the residual variance remains close to its baseline.

145 *Drift..* A gradually growing offset is distinguished from a static one by the
146 gap between the short- and long-horizon EWM means. We declare *drift* when
147 $|m_{\text{fast},i} - m_{\text{slow},i}|/\sigma_{\text{cond},i}$ exceeds a threshold, i.e. when the recent residual
148 mean is moving away from its longer-run value. We name this honestly: it
149 is a short-versus-long EWM mean-gap statistic, *not* a fitted regression slope.
150 It requires no stored window and no least-squares fit; it responds to the
151 direction and recency of residual movement rather than estimating a rate.

152 *Accuracy loss..* An inflated noise floor with an unchanged mean indicates
153 degraded measurement accuracy. We declare *accuracy loss* when the residual
154 variance v_i exceeds its baseline by a multiplicative factor while the slow
155 residual mean stays near zero.

156 *Freezing..* A transmitter stuck at its last value produces near-zero signal in-
157 crements while the conditional expectation continues to move. We declare
158 *freezing* when the EWM variance of the raw increments collapses below an
159 absolute floor `freeze_eps` and below a fraction `freeze_var_ratio` of the sig-
160 nal’s marginal variance, scaled by `freeze_abs_scale`. These three constants
161 govern how aggressively a quiet signal is judged frozen; their calibration is
162 examined in Section 5 (the freeze-on-quiescent ablation). A signal that satis-
163 fies none of the tests while flagged is labelled *normal* with respect to typing
164 (e.g. a peer signal pulled by cross-talk), and an unflagged signal is *normal*
165 by default.

166 3.4. Change-point graded suppression

167 A fault that arrives while the base detector is adapting to a genuine
168 regime shift can be either masked by the adaptation or mistaken for one.
169 The detector’s change-point machinery applies a graded suppression to the

170 typing decision around detected change points, controlled by a scale param-
 171 eter `suppress_scale`: at `scale = 1` no suppression is applied, the default
 172 `scale = 5` applies graded suppression, and `scale = ∞` suppresses any typ-
 173 ing within the change-point window entirely. This parameter trades off two
 174 failure modes that we quantify separately in Section 5: spurious type labels
 175 from coordinated shifts in weakly correlated signals (reduced by stronger sup-
 176 pression) against a masking dead-band for genuine faults co-occurring with a
 177 change point (widened by stronger suppression). We treat `suppress_scale`
 178 as the principal tuning knob of the composed system and report its effect
 179 rather than fixing it silently.

180 4. Experimental Setup

181 We evaluate the composed system along two complementary axes: con-
 182 trolled fault injection, which provides per-type ground truth, and a real fault
 183 deployment, which provides an external validity check. Because no public
 184 stream simultaneously labels all four fault types per signal and per timestep,
 185 neither axis alone suffices, and we triangulate the two. All evaluation artifacts
 186 referenced below are committed as CSV files under `examples/benchmarks/`.

187 4.1. Controlled fault injection

188 Following the injection-protocol precedent of Lai et al. (2021), we inject
 189 parameterized faults of each type—bias, drift, accuracy loss, and freezing—
 190 into a multivariate process with known ground truth, drawn from the Con-
 191 trolled Anomalies Time Series (CATS) benchmark (Fleith, 2023). Each typed
 192 fault is injected at controlled magnitude and onset onto a target signal,
 193 and recall is measured per type. We run $N = 85$ trials per type in the
 194 scaled controlled validation, and report Wilson 95% confidence intervals on
 195 every recall and precision estimate (`i7_scaled_per_type.csv`). To probe
 196 the dependence of bias recall on conditional coupling, we report bias re-
 197 call per target channel against that channel’s conditional standard deviation
 198 (`i7_scaled_bias_by_channel.csv`, $N = 17$ per channel). Freeze attribu-
 199 tion versus base detection is evaluated on the same channels, recording per-
 200 channel freeze and CDF detection rates and detection delays (`i7_freeze_latency.csv`).

201 4.2. Synthetic correlation control

202 To separate the two robustness mechanisms of the composed system,
 203 we generate synthetic multivariate streams with controlled pairwise corre-
 204 lation $\rho \in \{0.1, 0.3, 0.5, 0.7, 0.9\}$ and inject coordinated shifts of magnitude

205 $\{3, 6, 10\}$ conditional standard deviations co-occurring with a change point.
 206 For each cell we sweep the suppression scale `suppress_scale` $\in \{1, 5, \infty\}$
 207 and report the mean false-label rate over 5 seeds (`i7_regime_fp.csv`). The
 208 co-occurring-fault dead-band is measured by injecting a drift simultaneously
 209 with a change point and recording the detection magnitude at each sup-
 210 pression scale (`i7_deadband.csv`). The freeze-on-quiescent ablation injects
 211 no fault into a genuinely live but quiet signal (noise floor as a fraction of
 212 marginal standard deviation) and measures the false freeze-label rate under
 213 default, strict, and loose freeze constants (`i7_freeze_quiescent.csv`).

214 4.3. Real fault deployment

215 For external validity we use the Intel Berkeley Lab sensor deployment (Bodik
 216 et al., 2004), in which motes report temperature and humidity over a diurnal
 217 cycle. As a mote’s battery depletes (voltage below 2.3 V, held out as a physi-
 218 cal fault label), its temperature and humidity readings degrade to physically
 219 impossible values—temperatures of 100–122 °C and negative humidity. We
 220 treat the battery-voltage label as ground truth for a real fault, never expos-
 221 ing it to the detector or classifier. We sweep the detector’s mean-threshold
 222 operating point and report, pooled over motes, the healthy false-positive
 223 rate and the real-fault detection rate ($N_{\text{healthy}} \approx 1.52\text{M}$, $N_{\text{fault}} \approx 0.39\text{M}$;
 224 `i7_intel_lab_operating_points.csv`), together with the distribution of
 225 types assigned to detected real faults (`i7_intel_lab_fault_types.csv`).
 226 To establish that online adaptation is required rather than optional, we ad-
 227 ditionally fit a static model on a short prefix of the same stream and measure
 228 its false-alarm rate.

229 4.4. Metrics

230 For the controlled injections we report per-type recall (fraction of injected
 231 faults of a type that are detected) and cross-talk precision on normal peer
 232 signals. For the real deployment we report the healthy false-positive rate and
 233 the real-fault detection rate at each operating point. All proportions esti-
 234 mated from finite samples are reported with Wilson 95% confidence intervals.
 235 We do not report aggregate accuracy across types, because the type popula-
 236 tions are not equiprobable in either the injection design or the real stream,
 237 and an aggregate would obscure the per-type and per-coupling structure that
 238 is the substance of our characterization.

Table 1: Real battery-depletion fault detection on the Intel Berkeley Lab deployment, online adaptive detector, pooled over motes. Source: `i7_intel_lab_operating_points.csv` ($N_{\text{healthy}} = 1\,518\,828$, $N_{\text{fault}} = 387\,671$).

Mean threshold	Healthy FP rate	Real-fault detection
3.0	0.221	0.926
4.0	0.165	0.916
5.0	0.129	0.910
6.0	0.104	0.906
8.0	0.074	0.901

239 5. Results

240 5.1. Real-fault operating curve

241 On the Intel Berkeley Lab deployment, the composed system detects real
 242 battery-depletion faults at a high rate while keeping healthy false alarms low,
 243 and the operator can trade the two by moving a single mean-threshold op-
 244 erating point. Table 1 reports the pooled sweep over motes ($N_{\text{healthy}} \approx 1.52\text{M}$
 245 healthy samples, $N_{\text{fault}} \approx 0.39\text{M}$ fault samples; `i7_intel_lab_operating_points.csv`).
 246 At a mean threshold of 8.0 the system reaches a 7.4% healthy false-positive
 247 rate at 90.1% real-fault detection; lowering the threshold to 3.0 raises de-
 248 tection to 92.6% at the cost of a 22.1% false-positive rate. The detection
 249 rate is remarkably flat across the sweep (from 0.926 down to 0.901), so the
 250 operating point is chosen almost entirely by the tolerable false-alarm budget.

251 Online adaptation is mandatory on this stream, not a convenience. A
 252 static model fit on a short prefix of the same diurnal signal false-alarms at
 253 a rate of 0.97: the daily temperature and humidity cycle alone is enough to
 254 push a fixed model permanently out of distribution. The adaptive detector
 255 tracks the diurnal baseline and reserves its alarms for the battery-depletion
 256 excursions.

257 The types assigned to the detected real faults are dominated by freezing,
 258 as a stuck-at-degraded-value failure should be. Table 2 reports the type dis-
 259 tribution over detected real-fault samples (`i7_intel_lab_fault_types.csv`):
 260 93.2% are typed freezing, 5.1% drift, 1.7% bias, and 0.06% accuracy loss. The
 261 freezing label corresponds to the physical reality of a transmitter pinned at a
 262 corrupted reading as its battery fails, so the typing layer converts the detec-
 263 tor’s alarm into the maintenance-relevant diagnosis “replace the mote” rather
 264 than “investigate the process.”

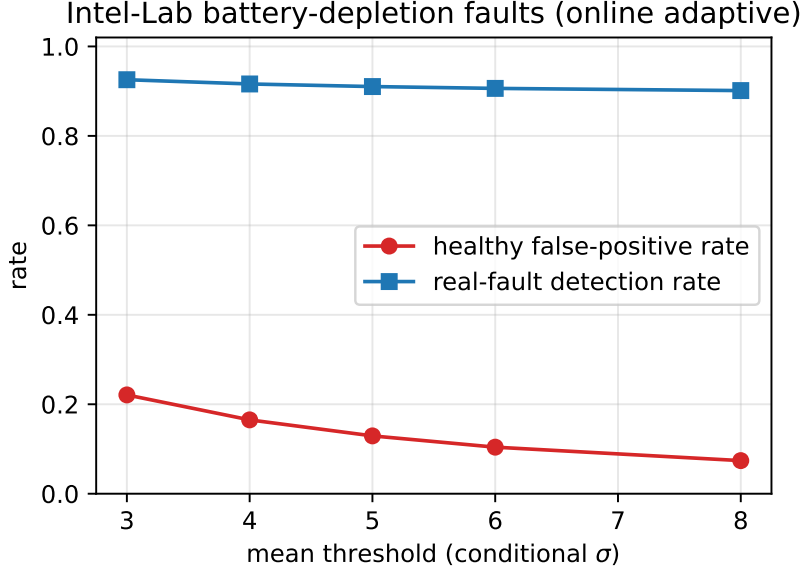


Figure 1: Operating curve on the Intel Berkeley Lab battery-depletion faults (online adaptive detector, pooled over motes). Raising the mean threshold trades a lower healthy false-positive rate against a marginal drop in real-fault detection. Source: `i7_intel_lab_operating_points.csv`.

265 The residual healthy false positives are themselves informative. Approx-
 266 imately 93% of healthy false positives are typed as drift, arising from the
 267 healthy diurnal ramps that the two-signal conditional model (temperature
 268 given humidity, and vice versa) cannot fully cancel. This is an honest limita-
 269 tion tied directly to the correlation-dependence we characterize in Section 5.4:
 270 with only two correlated signals available, slow shared ramps leak into the
 271 residual; a deployment with more correlated signals would suppress them
 272 further.

273 5.2. Per-type recall and the bias-coupling relationship

274 On the scaled controlled injections, the four types are recovered at sharply
 275 different rates. Table 3 reports per-type recall with Wilson 95% confidence
 276 intervals over $N = 85$ trials per type (`i7_scaled_per_type.csv`). Freezing is
 277 recovered perfectly (1.000, CI 0.957–1.000) and drift well (0.741), while accu-
 278 racy loss (0.659) and especially bias (0.518) are lower. Cross-talk precision on
 279 normal peer signals is high: of the signals labelled by typing, the normal-peer

Table 2: Fault types assigned to detected real faults on the Intel Berkeley Lab deployment. Source: `i7_intel_lab_fault_types.csv`.

Assigned type	Count	Fraction
freezing	578 721	0.932
drift	31 727	0.051
bias	10 395	0.017
accuracy loss	343	0.001

Table 3: Per-type recall on scaled controlled injections, $N = 85$ trials per type, Wilson 95% CI. Source: `i7_scaled_per_type.csv`.

Fault type	Recall	95% CI	N
freezing	1.000	[0.957, 1.000]	85
drift	0.741	[0.639, 0.822]	85
accuracy loss	0.659	[0.553, 0.751]	85
bias	0.518	[0.413, 0.621]	85

label is correct with precision ≈ 0.91 (`i7_scaled_normal_precision.csv`), so coupled peers are rarely misattributed a fault.

The aggregate bias recall of 0.518 is not an unexplained weakness; it is the mean of two distinct regimes. Table 4 reports bias recall per target channel against that channel’s conditional standard deviation σ (`i7_scaled_bias_by_channel.csv`). Bias recall is monotone in σ : on strongly coupled channels (small conditional σ , because peers explain most of the variance) the recall is near-perfect (0.941 at $\sigma = 0.216$, 1.000 at $\sigma = 0.480$), whereas on weakly coupled channels (large conditional σ) it collapses (0.353, 0.235, down to 0.059 at $\sigma = 1.471$). This is the key finding of the characterization: the interpretable conditional model *absorbs* a constant offset on a strongly coupled signal, because the conditional mean tracks the offset through the peers, while it *isolates* the same offset on a weakly coupled signal. The aggregate 0.52 is thus the average of an “isolable” regime and an “absorbed” regime, not a uniformly mediocre detector.

5.3. Freezing is type attribution, not detection capability

We initially expected freezing to be a detection blind spot of the base detector, on the reasoning that a signal stuck at a value within its nor-

Table 4: Bias recall is monotone in conditional standard deviation σ : offsets are absorbed on strongly coupled (small- σ) channels and isolated on weakly coupled (large- σ) channels. $N = 17$ per channel, Wilson 95% CI. Source: `i7_scaled_bias_by_channel.csv`.

Channel	Conditional σ	Bias recall	95% CI	N
bed2	0.216	0.941	[0.73, 0.99]	17
bed1	0.480	1.000	[0.816, 1.00]	17
bso1	1.439	0.353	[0.173, 0.587]	17
bfo2	1.443	0.235	[0.096, 0.473]	17
cso1	1.471	0.059	[0.01, 0.27]	17

Table 5: Freezing detection: the base CDF detector catches every injected freeze on every channel, usually faster than the fixed 25-step freeze test. Source: `i7_freeze_latency.csv` ($N = 17$ per channel).

Channel	Cond. σ	CDF detect	CDF median delay	Freeze earlier	CDF never
bed2	0.216	1.0	28	0.529	0.0
bed1	0.480	1.0	9	0.294	0.0
bso1	1.439	1.0	17	0.412	0.0
bfo2	1.443	1.0	3	0.000	0.0
cso1	1.471	1.0	2	0.000	0.0

mal range would not trigger a conditional-CDF alarm. This claim is refuted by our own validation. Table 5 reports, per channel, the base CDF detection rate, the dedicated freeze-test detection rate, their median detection delays, and the fraction of trials in which the freeze test fired earlier (`i7_freeze_latency.csv`). The base CDF detector catches *every* injected freeze on *every* channel (`cdf_detect_rate` = 1.0, `frac_cdf_never` = 0.00 throughout), because the conditional expectation keeps moving while the frozen signal does not, driving the residual out of bounds. Moreover, the CDF detector is usually *faster*: its median delay ranges from 2 to 28 steps, and the freeze test—whose fixed 25-step confirmation window sets its median delay at exactly 25—fires earlier in only about half the trials, and only on the most isolated channel (bed2, `frac_freeze_earlier` = 0.529); on the most coupled channels it never beats the CDF detector (0.0).

The conclusion is that the freeze test adds type *attribution*—distinguishing a stuck sensor from a genuine process excursion—rather than detection capa-

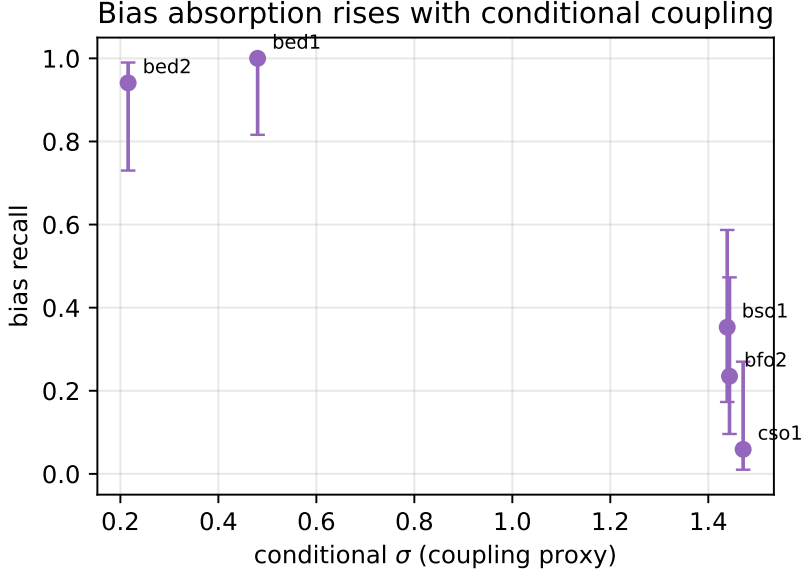


Figure 2: Bias recall is monotone in conditional σ : a constant offset is isolated on weakly coupled (small- σ) channels and absorbed by the conditional mean on strongly coupled (large- σ) channels. Error bars are Wilson 95% intervals. Source: `i7_scaled_bias_by_channel.csv`.

313 bility or speed. The theoretical blind spot, in which the conditional standard
 314 deviation collapses to zero and the residual cannot grow, is an edge case that
 315 does not arise in the real or injected data we examined. On the real deploy-
 316 ment this attribution is exactly what types the battery-depletion faults as
 317 freezing (Section 5.1), turning a detection into a maintenance label.

318 5.4. Regime robustness comes from suppression, not residual cancellation

319 We also expected the conditional residual to cancel coordinated shifts—
 320 a fault arriving as a shared shift across correlated signals would, on this
 321 reasoning, leave the conditional residual unchanged and so produce no spu-
 322 rious type label, independent of the change-point suppression. This claim
 323 is also refuted: residual cancellation (mechanism 1) is strongly correlation-
 324 dependent, and the robustness of the composed system instead comes from
 325 graded change-point suppression (mechanism 2). Table 6 reports the false-
 326 label rate across correlation ρ , coordinated-shift magnitude, and suppres-
 327 sion scale (`i7_regime_fp.csv`, 5 seeds per cell). With suppression disabled
 328 (`scale = 1`), a $\geq 6\sigma$ coordinated shift at low correlation ($\rho = 0.1$) produces

Table 6: False-label rate under coordinated shifts co-occurring with a change point, by correlation ρ , shift magnitude, and suppression scale (mean over 5 seeds). Residual cancellation alone (**scale** = 1) fails at low correlation; graded suppression (**scale** = 5) yields 0.0 everywhere. Source: `i7_regime_fp.csv`.

ρ	Shift (σ)	scale =1	scale =5	scale = ∞
0.1	3	0.000	0.000	0.000
0.1	6	0.199	0.000	0.000
0.1	10	0.201	0.000	0.000
0.3	3	0.000	0.000	0.000
0.3	6	0.021	0.000	0.000
0.3	10	0.198	0.000	0.000
0.5	3	0.000	0.000	0.000
0.5	6	0.000	0.000	0.000
0.5	10	0.112	0.000	0.000
0.7	3–10	0.000	0.000	0.000
0.9	3–10	0.000	0.000	0.000

about 20% false labels (0.199 at 6σ , 0.201 at 10σ); the boundary recedes as ρ increases and is gone by $\rho \geq 0.7$. By contrast, the default graded suppression (**scale** = 5) drives the false-label rate to exactly 0.0 in every (ρ, shift) cell, as does full suppression (**scale** = ∞).

The adapt-into-drift robustness of the composed system therefore rests on mechanism 2, the graded change-point suppression, and not on the conditional-residual cancellation we originally credited. This matters operationally: the suppression scale is the parameter the operator must set, and it has a measurable cost, quantified next.

5.5. The cost of suppression: dead-band and freeze-on-quiescent ablations

Suppression suppresses real faults too. Table 7 reports the detection of a genuine drift that co-occurs with a change point, as a function of suppression scale (`i7_deadband.csv`, 5 seeds). With no suppression (**scale** = 1) the drift is detected once it reaches 5.2σ ; at the default **scale** = 5 detection is delayed until 17.6σ ; at **scale** = ∞ the drift is never detected. The default suppression thus opens a masking dead-band of roughly $[3\sigma, 15\sigma]$ for faults that happen to coincide with a change point—the measured price of the robustness reported in Section 5.4.

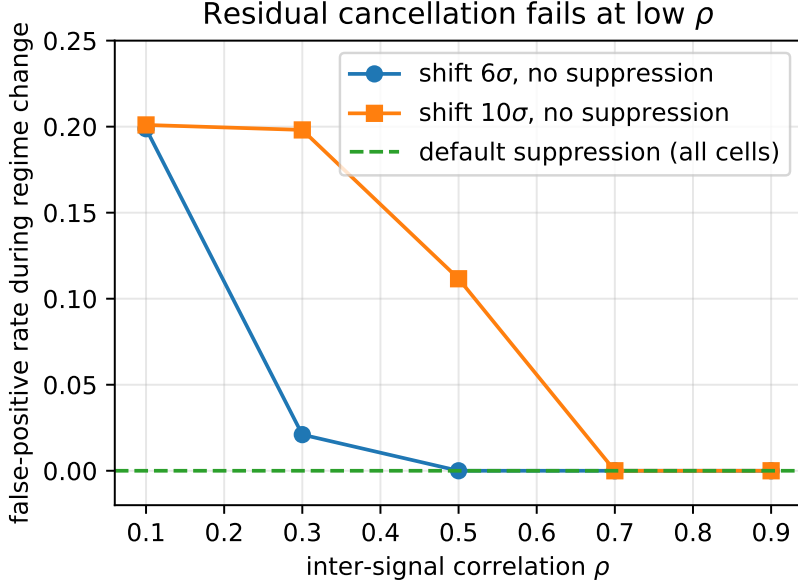


Figure 3: Without change-point suppression, conditional-residual cancellation (mechanism 1) fails at low correlation: a large coordinated shift produces false fault labels until ρ is high. The default graded suppression (mechanism 2) drives the rate to zero in every cell. Source: `i7_regime_fp.csv`.

347 The freeze test similarly requires calibration on steady-state signals. Ta-
 348 ble 8 reports the false freeze-label rate on a genuinely live but very quiet
 349 signal (noise at 0.5% of the marginal standard deviation) under three freeze-
 350 constant settings (`i7_freeze_quiescent.csv`, 5 seeds). With the default
 351 constants, such a signal is falsely labelled frozen 72% of the time; the loose
 352 constants make it worse ($\approx 88\%$); the strict constants (`freeze_eps` = 10^{-4} ,
 353 `freeze_var_ratio` = 5×10^{-3} , `freeze_abs_scale` = 5) eliminate the false
 354 label entirely (0.0). We therefore recommend the strict constants, or per-
 355 signal calibration, for deployments containing steady-state signals.

356 6. Discussion

357 The characterization in Section 5 maps the operating envelope of the
 358 composed system honestly. We read each apparent limitation as a quantified
 359 property that an operator can plan around, rather than a defect to be hidden.

Table 7: Co-occurring-fault dead-band: detection magnitude of a drift co-occurring with a change point, by suppression scale (5 seeds). Source: `i7_deadband.csv`.

Suppression scale	Detected	Median delay	Magnitude at detection (σ)
1.0	yes	26	5.2
5.0	yes	88	17.6
∞	no	—	—

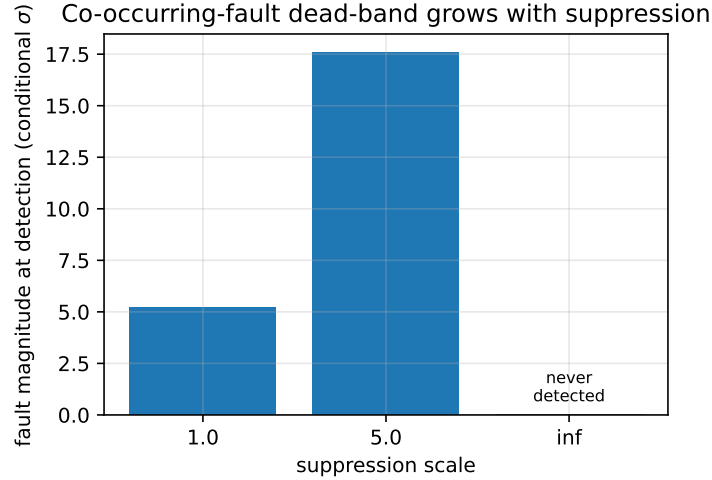


Figure 4: The co-occurring-fault dead-band: a drift co-occurring with a change point must grow from 5.2σ (no suppression) to 17.6σ (default suppression) before detection, and is never detected under full suppression. Source: `i7_deadband.csv`.

360 *Bias absorption is a property of the interpretable model, not a bug.* The
361 monotone dependence of bias recall on conditional standard deviation (Ta-
362 ble 4) follows directly from what makes the base detector interpretable. Be-
363 cause the conditional mean of a strongly coupled signal tracks its peers, a
364 constant offset injected on that signal is partly explained away by the condi-
365 tional model, and the residual—hence the typing—does not see it. The same
366 mechanism that yields tight, peer-informed operating limits also absorbs off-
367 sets on those signals. The practical reading is that bias is reliably typed
368 where it is observable (weakly coupled signals show recall up to 0.941) and is
369 absorbed where the model is confident enough to overrule it; the aggregate
370 0.518 is the average of these two regimes, not a uniform failure. An oper-
371 ator who needs bias coverage on a strongly coupled signal should monitor

Table 8: Freeze-on-quiescent ablation: false freeze-label rate on a live but quiet signal (noise 0.5% of marginal std), by freeze-constant setting (5 seeds). Source: `i7_freeze_quiescent.csv`.

Freeze constants	<code>freeze_eps</code>	<code>var_ratio</code> / <code>abs_scale</code>	False freeze rate
default	1×10^{-3}	1×10^{-2} / 20	0.717
loose	5×10^{-3}	5×10^{-2} / 50	0.887
strict	1×10^{-4}	5×10^{-3} / 5	0.000

372 that signal’s raw calibration directly, a check the conditional model is, by
 373 construction, not designed to provide.

374 *Two-signal cancellation limits the real-deployment false-positive floor..* On
 375 the Intel Lab stream the residual healthy false positives are $\approx 93\%$ drift,
 376 traced to diurnal ramps that a two-signal conditional model cannot fully
 377 cancel. This is the same correlation-dependence quantified in Table 6: with
 378 few correlated signals, slow shared movement leaks into the residual. The
 379 limitation is therefore structural and predictable—it improves with the num-
 380 ber of correlated signals available—rather than a tuning failure. We report
 381 the 7.4% floor at 90% detection as the achievable operating point on a two-
 382 signal deployment, with the explicit expectation that richer instrumentation
 383 would lower it.

384 *The dead-band is the priced cost of adaptation robustness..* Because robust-
 385 ness to faults arriving during adaptation comes from graded change-point
 386 suppression and not from residual cancellation (Section 5.4), it carries a
 387 cost: the masking dead-band of roughly $[3\sigma, 15\sigma]$ for faults co-occurring with
 388 a change point (Table 7). We present this as an explicit trade-off governed
 389 by a single parameter, `suppress_scale`. A deployment that fears spurious
 390 labels from coordinated shifts should keep the default suppression; one that
 391 fears missing a fault during a regime change should lower it, accepting the
 392 false-label rate of Table 6. The point is that the trade-off is measured and
 393 tunable, not silent.

394 *Threshold and freeze-constant calibration are deployment decisions..* The operating-
 395 point sweep (Table 1) and the freeze-on-quiescent ablation (Table 8) both
 396 show that the system’s behavior is set by a small number of interpretable con-
 397 stants. The mean threshold selects the false-alarm/detection trade-off; the

398 freeze constants determine how aggressively a quiet signal is judged frozen,
 399 with the strict setting eliminating false freeze labels on a live-but-quiet signal
 400 that the default mislabels 72% of the time. We recommend the strict freeze
 401 constants or per-signal calibration for steady-state signals, and we expose the
 402 threshold as the operator’s primary control. These are calibration decisions
 403 with quantified consequences, not hidden defaults.

404 *Data availability constrains the evaluation to triangulation..* No public stream
 405 simultaneously labels all four fault types per signal and per timestep, which
 406 is why we combine controlled injection (per-type ground truth, but syn-
 407 thetic) with a real fault deployment (genuine faults, but a single physical
 408 failure mode with a held-out label). Neither alone would be conclusive: the
 409 injections could overstate performance on idealized faults, and the real de-
 410 ployment exercises mainly the freezing signature. Read together, the injec-
 411 tions establish the per-type envelope and the refuted claims, while the real
 412 deployment confirms that the envelope holds on genuine faults at a usable
 413 operating point. We regard this triangulation as the appropriate evidentiary
 414 standard given the data that exist, and we flag the absence of a fully labelled
 415 multi-type stream as the principal external constraint on stronger claims.

416 7. Conclusion

417 We presented a streaming sensor-fault classifier that closes the detection-
 418 to-action gap of a self-supervised online anomaly detector (Wadinger and
 419 Kvasnica, 2023) by attributing each flagged signal to a maintenance-relevant
 420 fault type—bias, drift, accuracy loss, or freezing—from the detector’s own
 421 conditional residuals, in $O(1)$ memory and with no history buffers. The
 422 fault taxonomy is the established one (Sharma et al., 2010; Ni et al., 2009);
 423 the contribution is its streaming, conditional-residual realization composed
 424 directly with the detector, so that type attribution is available at the same
 425 latency and operational footprint as detection.

426 The headline result is practical. On real battery-depletion faults in the
 427 Intel Berkeley Lab deployment (Bodik et al., 2004), the composed system
 428 reaches 90% real-fault detection at a 7.4% healthy false-positive rate and
 429 types the detected faults predominantly as freezing (0.93), the correct main-
 430 tenance diagnosis for a transmitter stuck at a degraded value. Online adap-
 431 tation is shown to be mandatory: a static prefix model false-alarms at 0.97
 432 on the same diurnal stream.

433 Around this result we characterized the operating envelope rather than
434 overstating it. Freezing and drift are recovered with recall 1.000 and 0.741 on
435 controlled injections, while bias recall is monotone in conditional coupling—
436 absorbed on strongly coupled signals and isolated on weakly coupled ones—so
437 that the aggregate 0.518 is the mean of two explained regimes. We reported
438 two refuted claims as such: freezing is not a detection blind spot of the
439 base detector (which catches every injected freeze, usually faster than the
440 dedicated test, so the freeze test contributes type attribution rather than
441 detection), and adaptation robustness arises from graded change-point sup-
442 pression rather than from conditional-residual cancellation. The suppression
443 carries a measured masking dead-band of roughly $[3\sigma, 15\sigma]$ for co-occurring
444 faults, and the freeze test requires strict constants or per-signal calibration
445 on steady-state signals.

446 The scope is correspondingly honest. The real-deployment evidence exer-
447 cises principally the freezing signature, the two-signal conditional model sets
448 a structural false-positive floor that richer instrumentation would lower, and
449 per-type ground truth is available only through injection because no public
450 stream labels all four fault types per signal and per timestep. Within that
451 scope, the system delivers a deployable typing layer that turns an adaptive
452 detector’s alarm into an actionable maintenance label, with an operating
453 envelope that is mapped rather than assumed.

454 **Additional information**

455 The classifier and all evaluation artifacts are openly accessible on GitHub
456 at the following URL: [https://github.com/MarekWadinger/online_outlier_](https://github.com/MarekWadinger/online_outlier_detection)
457 [detection](https://github.com/MarekWadinger/online_outlier_detection).

458 **CRediT authorship contribution statement**

459 **Marek Wadinger:** Conceptualization; Data curation; Formal analysis;
460 Investigation; Methodology; Software; Validation; Visualization; Writing -
461 original draft; and Writing - review & editing. **Michal Kvasnica:** Concep-
462 tualization; Funding acquisition; Project administration; Resources; Super-
463 vision; Validation.

464 Declaration of Competing Interest

465 The authors declare that they have no known competing financial inter-
466 ests or personal relationships that could have appeared to influence the work
467 reported in this paper.

468 Acknowledgements

469 The authors gratefully acknowledge the contribution of the Slovak Re-
470 search and Development Agency under the project APVV-20-0261. The au-
471 thors gratefully acknowledge the contribution of the Scientific Grant Agency
472 of the Slovak Republic under the grant 1/0490/23. This paper was funded
473 by the European Union under Horizon Europe Grant Agreement number
474 101079342 (Fostering Opportunities Towards Slovak Excellence in Advanced
475 Control for Smart Industries).

476 References

- 477 Bodik, P., Hong, W., Guestrin, C., Madden, S., Paskin, M., Thibaux, R.,
478 2004. Intel berkeley research lab sensor data. [https://db.csail.mit.](https://db.csail.mit.edu/labdata/labdata.html)
479 [edu/labdata/labdata.html](https://db.csail.mit.edu/labdata/labdata.html).
- 480 Fleith, P., 2023. Controlled anomalies time series (cats) dataset. doi:10.
481 5281/zenodo.8338435. solenix Engineering GmbH; CC-BY-4.0.
- 482 Lai, K.H., Zha, D., Xu, J., Zhao, Y., Wang, G., Hu, X., 2021. Revisiting time
483 series outlier detection: Definitions and benchmarks, in: Advances in Neu-
484 ral Information Processing Systems (NeurIPS) Datasets and Benchmarks
485 Track.
- 486 Ni, K., Ramanathan, N., Chehade, M.N.H., Balzano, L., Nair, S., Zahedi, S.,
487 Kohler, E., Pottie, G., Hansen, M., Srivastava, M., 2009. Sensor network
488 data fault types, pp. 1–29. doi:10.1145/1525856.1525863.
- 489 Sharma, A.B., Golubchik, L., Govindan, R., 2010. Sensor faults: Detection
490 methods and prevalence in real-world datasets. ACM Transactions on
491 Sensor Networks 6, 1–39. doi:10.1145/1754414.1754419.
- 492 Wadinger, M., Kvasnica, M., 2023. Real-time outlier detection with dynamic
493 process limits, in: 2023 24th International Conference on Process Control
494 (PC), pp. 138–143. doi:10.1109/PC58330.2023.10217717.